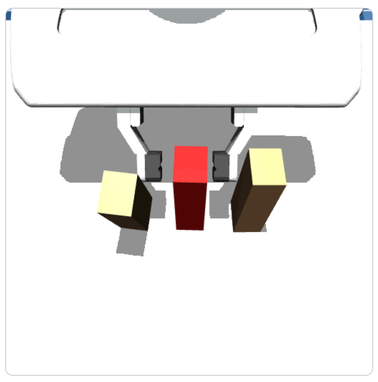
Do World Models Branch Where Physics Does? An Intuitive-Physics Test for Safe Manipulation

Alikasim Budhwani · Soo Jeon
Mechanical and Mechatronics Engineering, University of Waterloo · a2budhwa@uwaterloo.ca

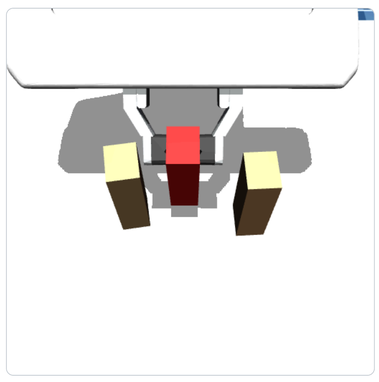
A world model can predict well on average, even with the right amount of uncertainty, and still miss where physics branches. A safety check needs the second.

1 The question

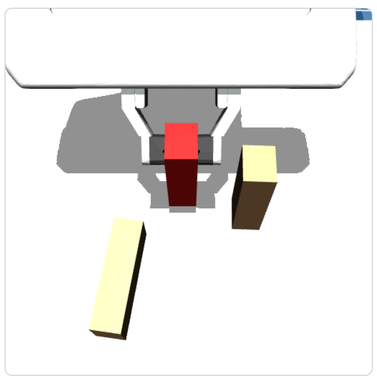
Before each action chunk, a robot should ask: **could my own small errors change what happens?** Picking the middle block, the same command lifts it cleanly or tips a neighbour, depending on a few millimetres of perception and execution error.



the chunk starts



31 of 64 futures: neighbour stays

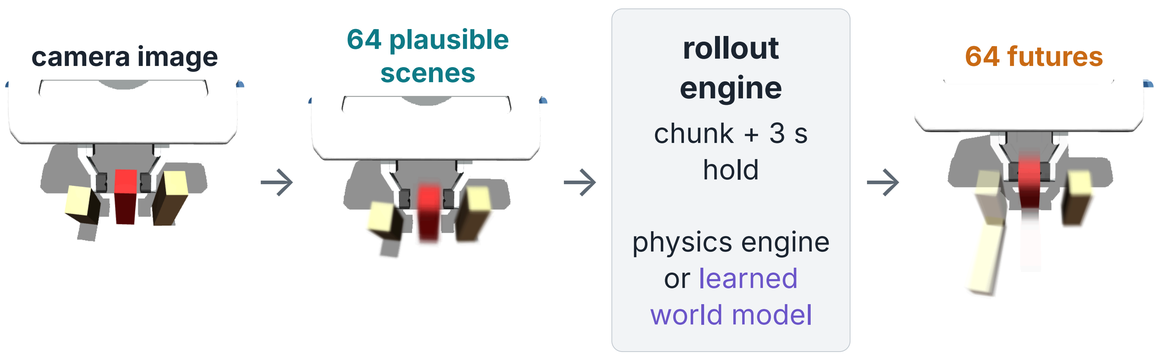


33 of 64 futures: neighbour topples

The physics is deterministic: all uncertainty is the robot's own. A state is a **fork** if the outcome splits over futures sampled from that uncertainty.

2 The fork probe

Like the human intuitive physics engine: a few rough simulations over an uncertain scene; disagreement means precarious.



- Samples the robot's **measured** errors: perception 2.5 mm, execution about 3 mm.
- Flags the chunk if the 64 endings are **bimodal**, not merely different.
- No task labels, no failure data, no thresholds fit to labels.

3 With a physics engine, it works

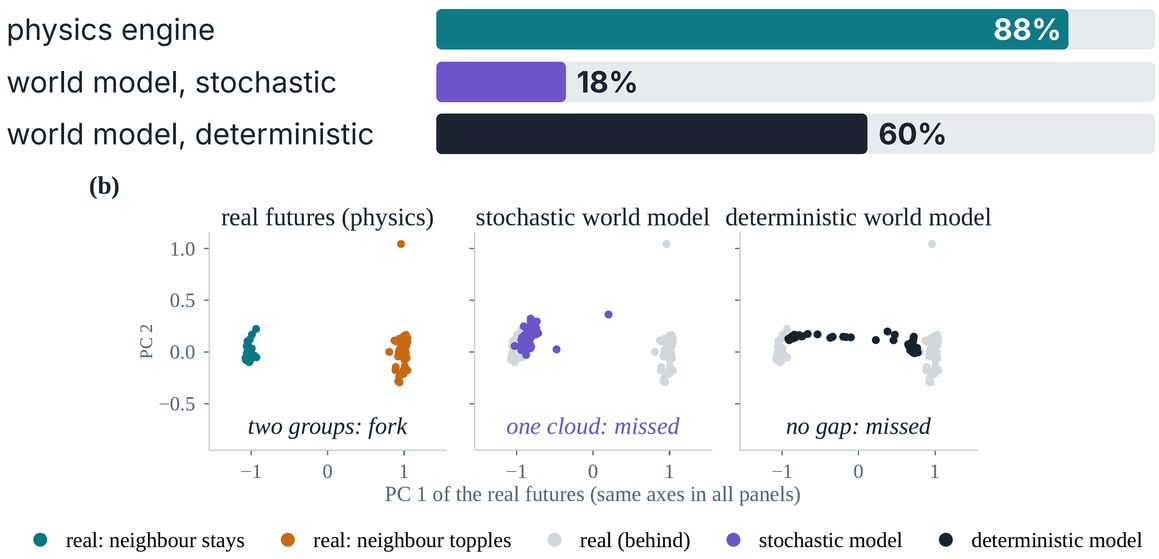
88% of forks caught with **7%** false alarms on unseen layouts, from one camera image (test set C, 156 states; 95% CI 77–94%)



Sampling perception error matters: it is as large as execution error. This is the bar a learned world model must reach.

4 Learned world models miss the branch

Replace the engine with a world model predicting in DINOv2 feature space. Both models below predict **3–4× better than a no-motion baseline**.



Right amount, wrong shape. The stochastic model's spread matches reality (0.98×), but only 15% lies along the branch (real: 57%), and it is barely larger on forks than on safe states (1.3× vs 4.8×). Five more generative variants catch at most 3%.

5 What closes the gap

Uncertain starts, deterministic dynamics, as in the intuitive physics engine: perturb the start, roll it out without noise. **Oversample rare branches**: find each training scene's rare branch with the same split test, no labels.



False alarms 7–11%. A generative model given the same uncertain starts still catches 0%: noise in the dynamics erases the forks.

6 Takeaways

Measure branching fidelity

Average error and calibrated spread do not show whether rollouts split where physics splits.

Put uncertainty in the start

Deterministic dynamics from perturbed starts keep the branch; per-step noise smears it.

Train for rare branches

Oversampling each scene's rare branch, found without labels, lifts catch rates.

A fork is not a failure

Pair the flag with a harm signal, such as whether the outcome can be undone, before acting.

Limitations: simulation, one three-block task; the ground truth uses the same error model the probe samples; single-seed, small world models.